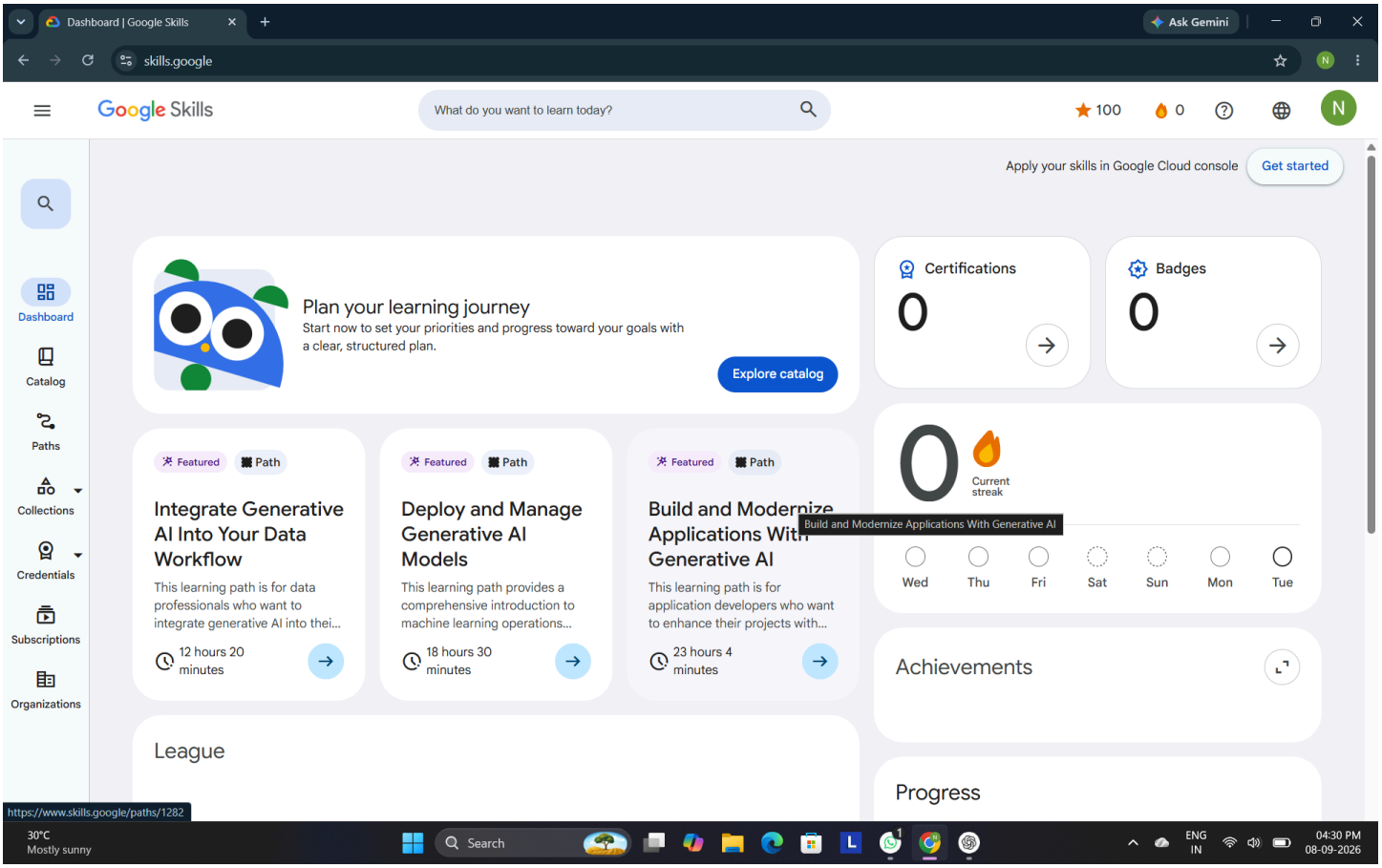
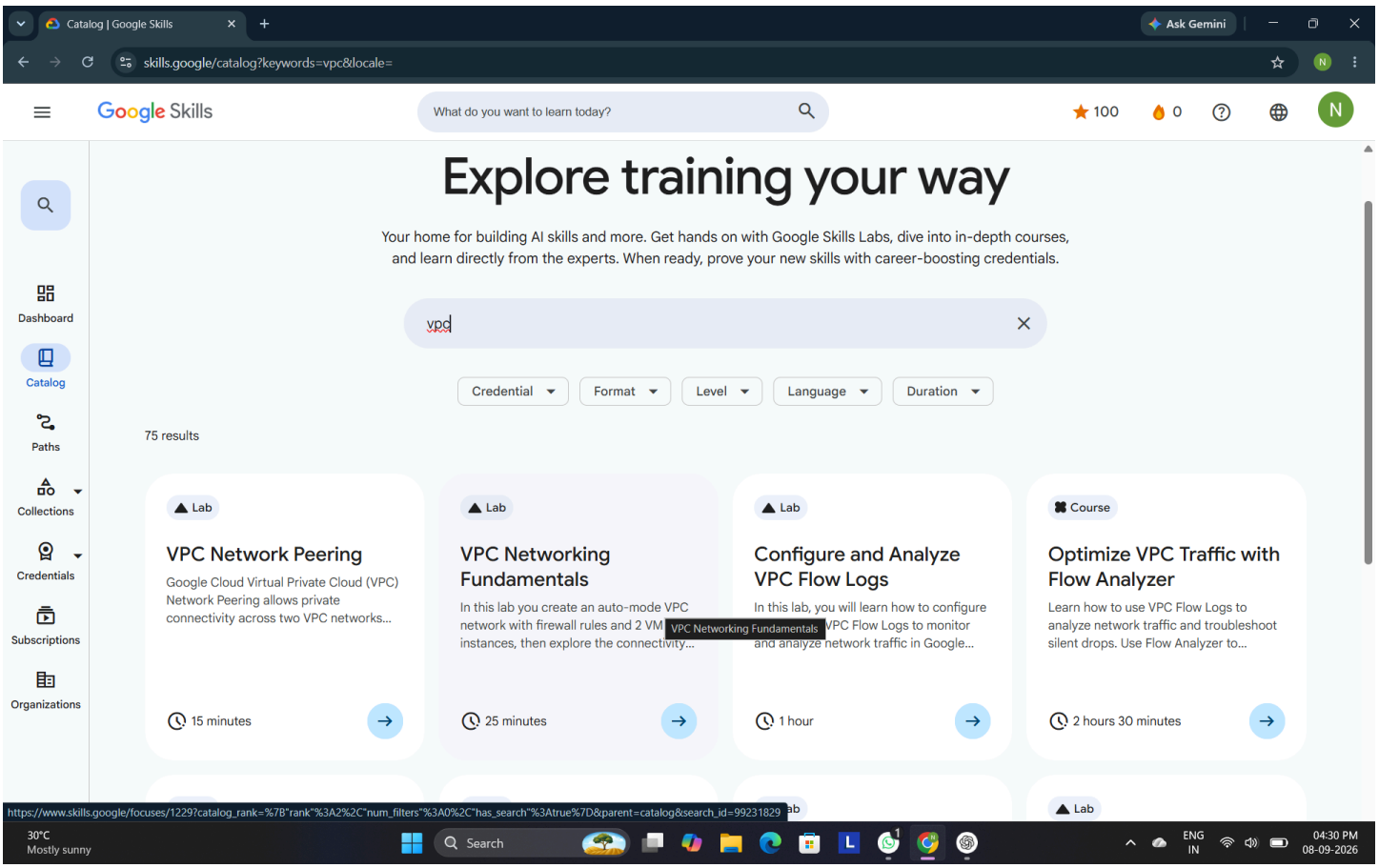
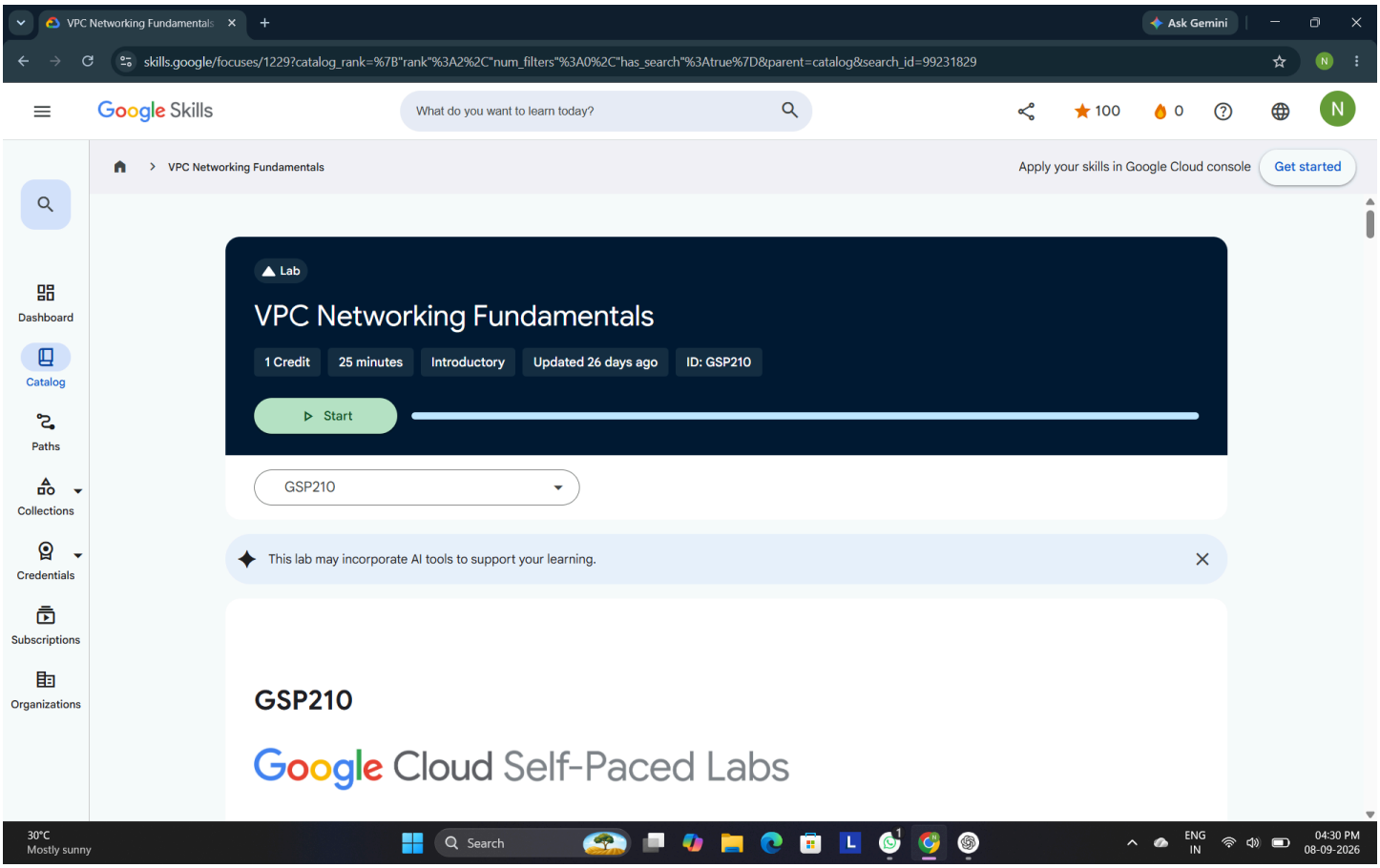
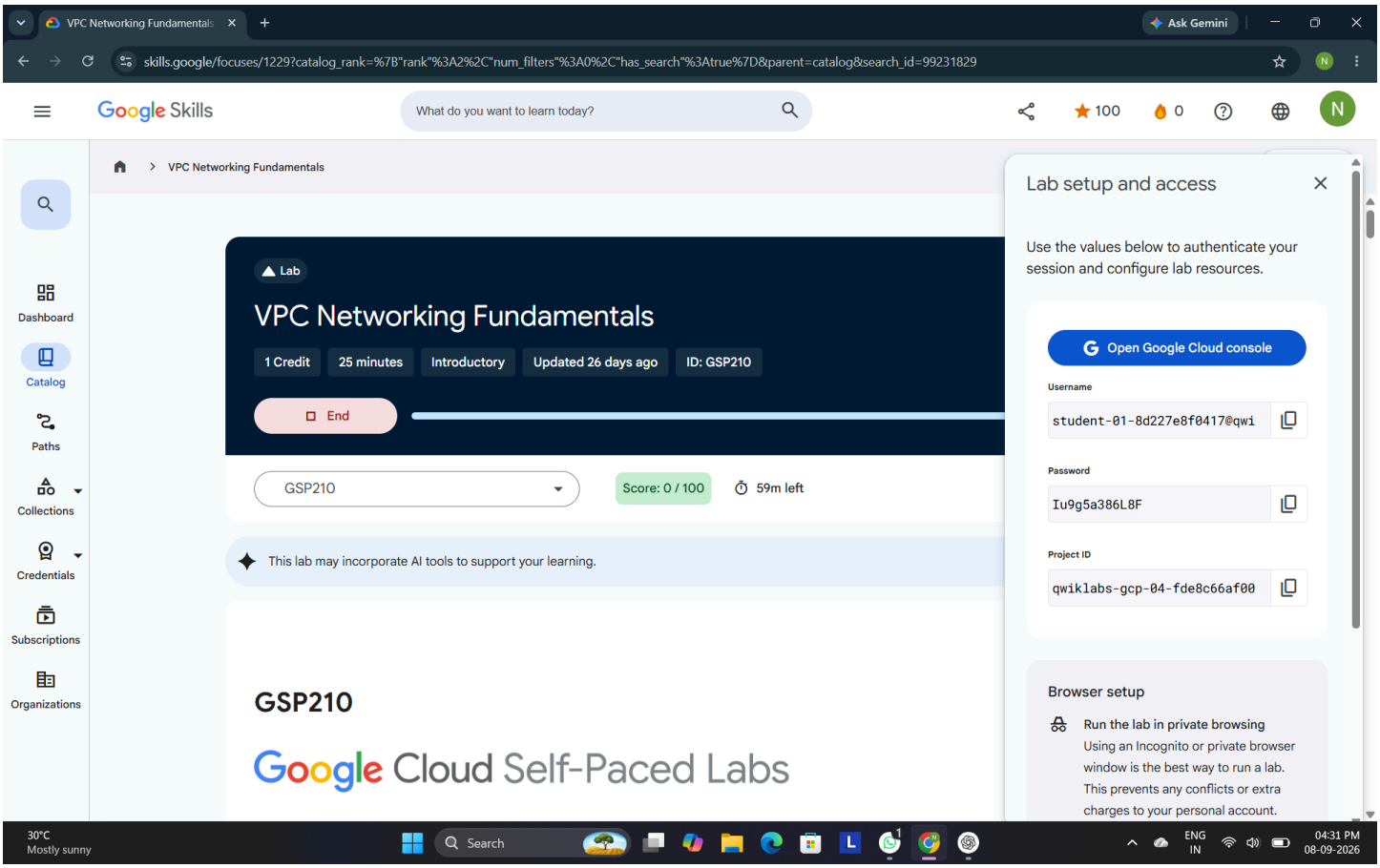
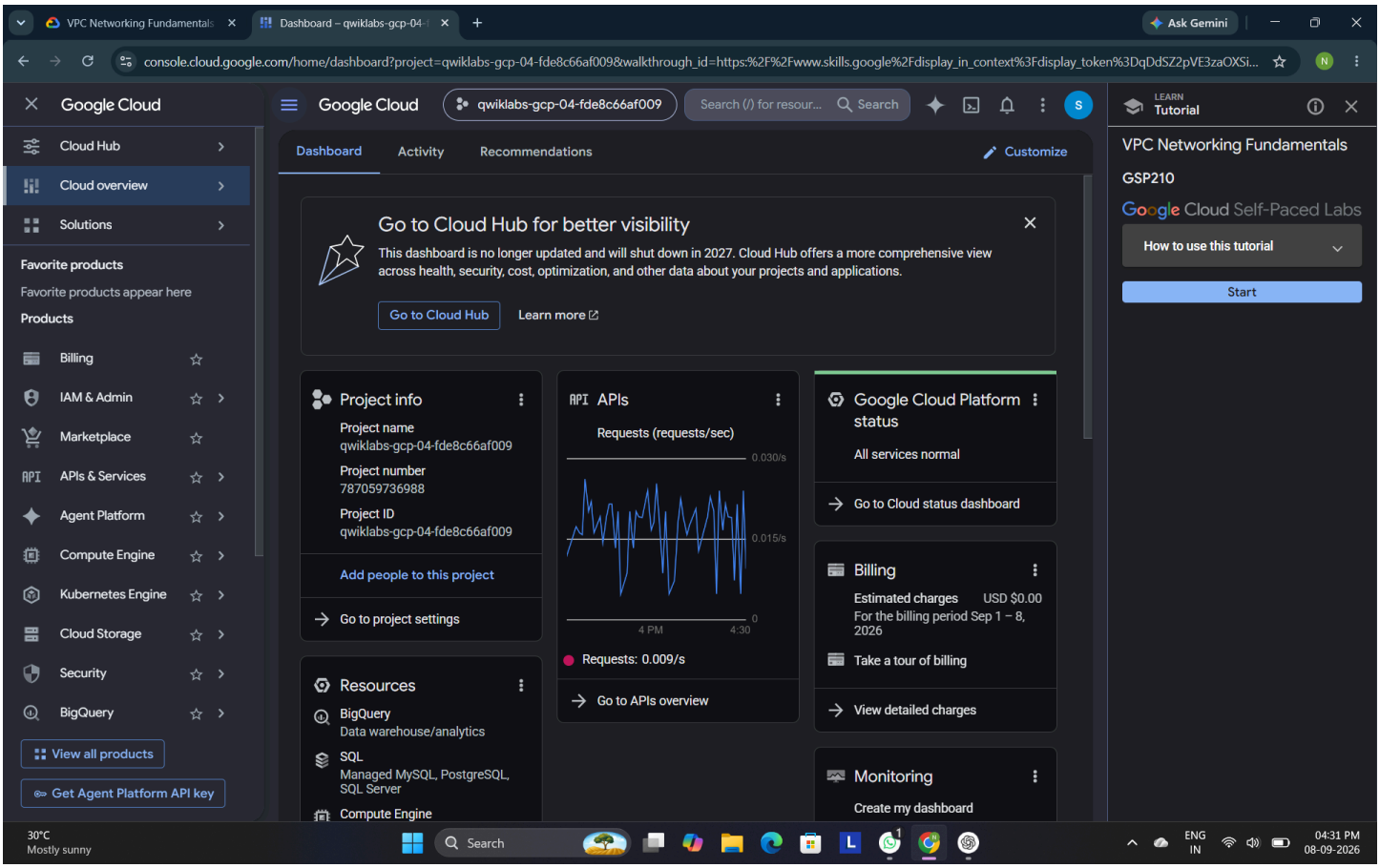
VPC NETWORKS AND VM CREATION

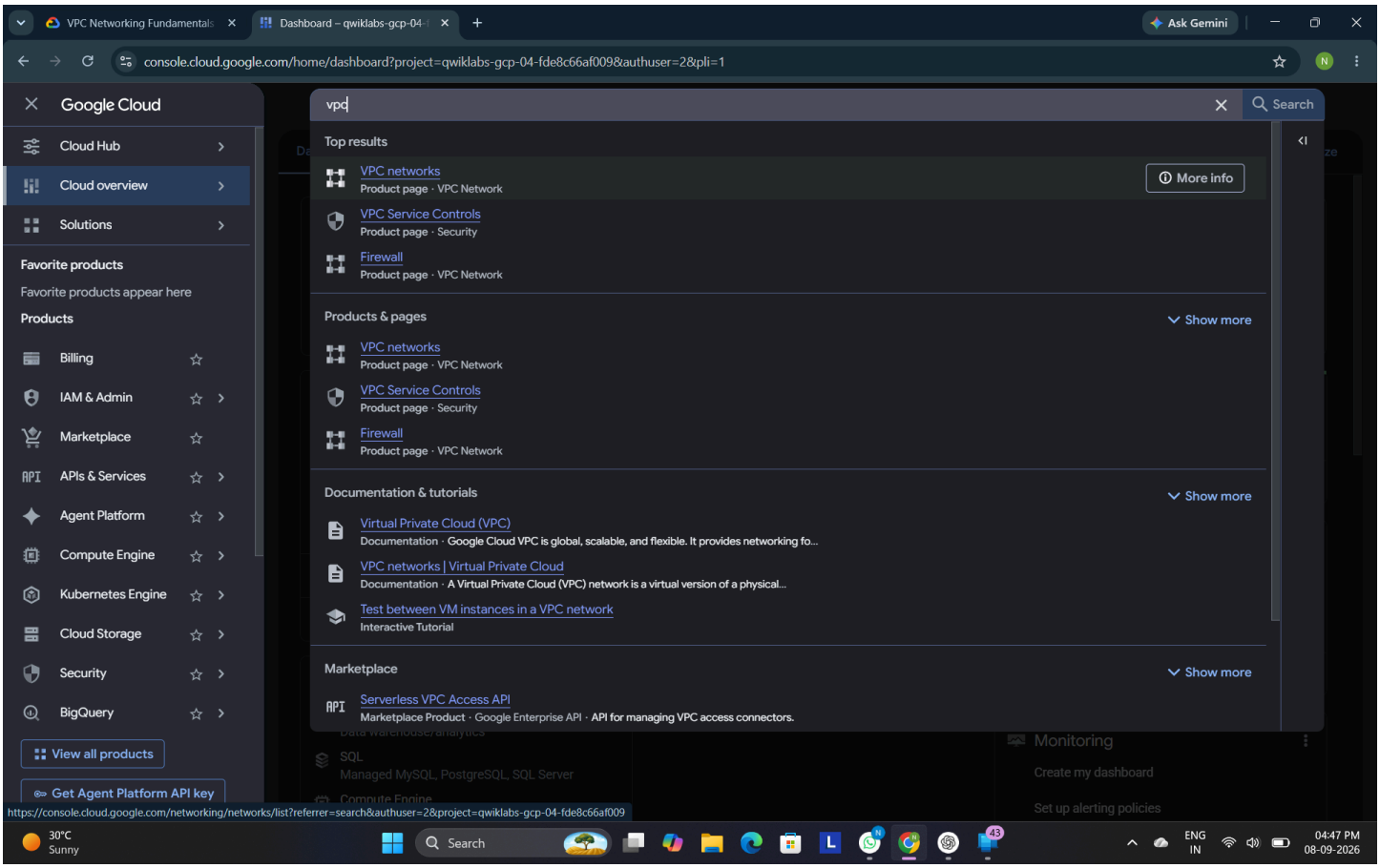


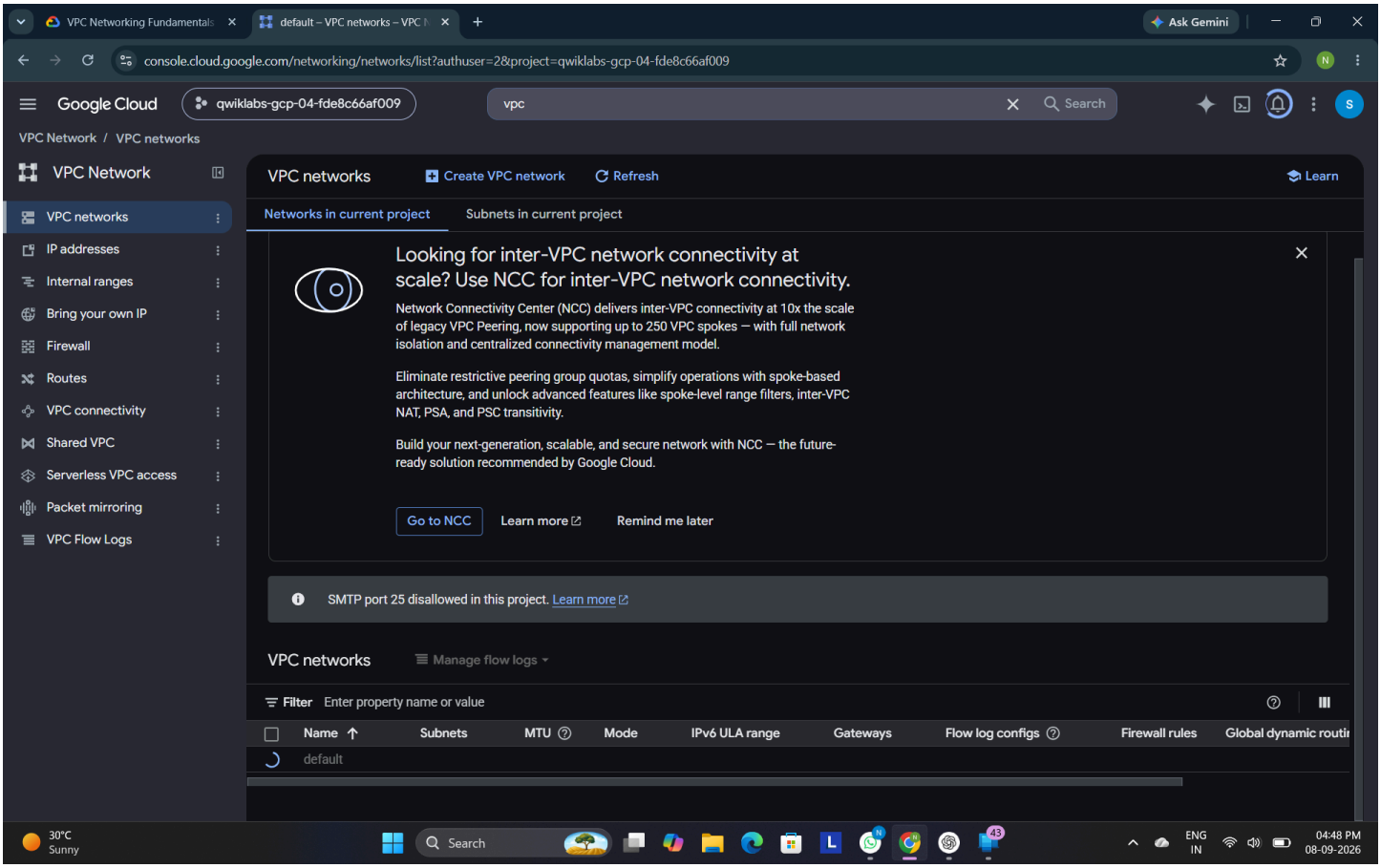


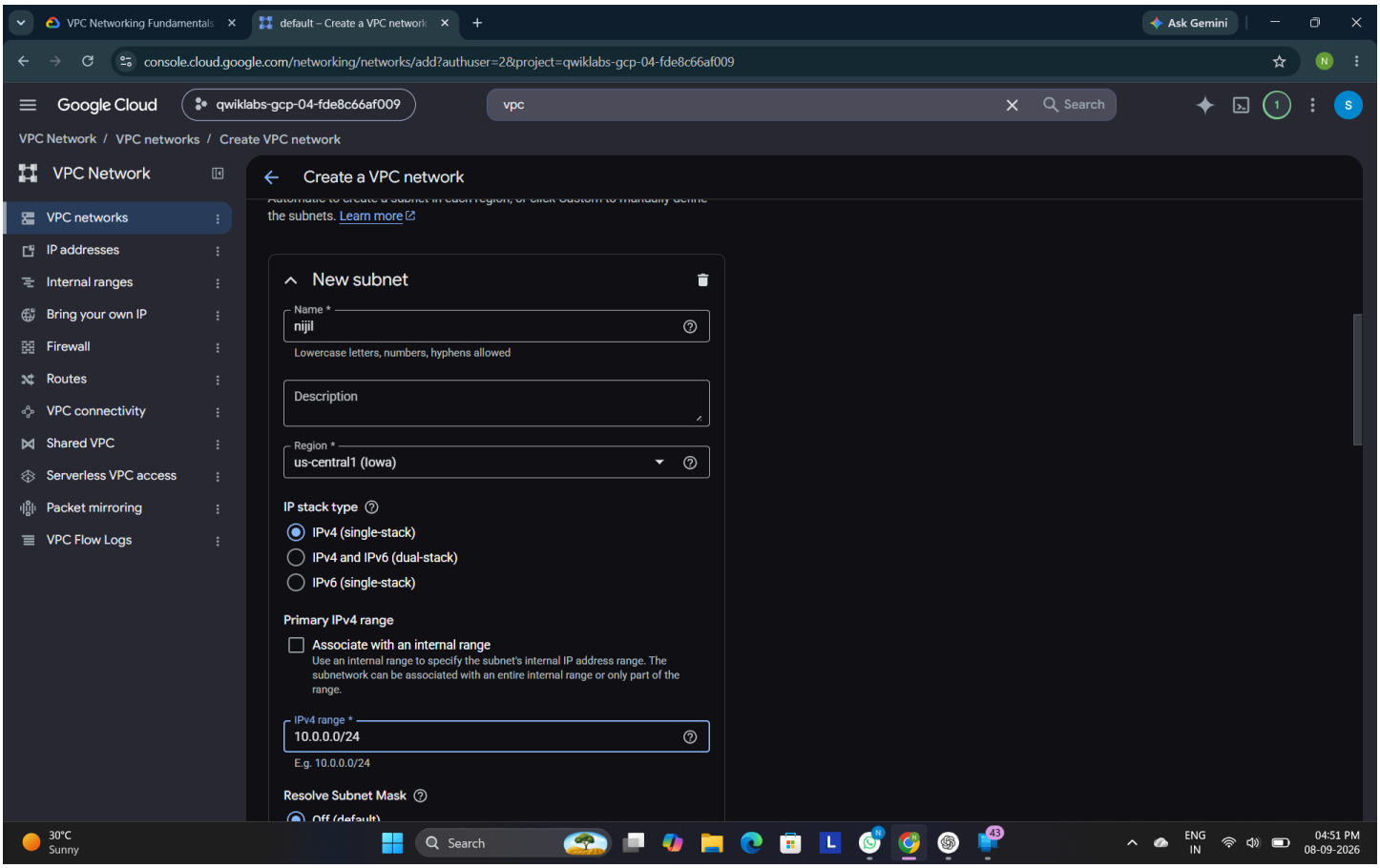

VPC Networking Fundamentals
Ask Gemini
skills.google/focuses/1229?catalog_rank=%7B"rank"%3A2%2C"num_filters"%3A0%2C"has_search"%3Atrue%7D&parent=catalog&search_id=99231829
Google Skills
What do you want to learn today?
100
0
VPC Networking Fundamentals
Apply your skills in Google Cloud console
Get started
Dashboard
Catalog
Paths
Collections
Credentials
Subscriptions
Organizations
Lab
VPC Networking Fundamentals
1 Credit
25 minutes
Introductory
Updated 26 days ago
ID: GSP210
Start
GSP210
This lab may incorporate AI tools to support your learning.
GSP210
Google Cloud Self-Paced Labs
30°C
Mostly sunny
Search
ENG
IN
04:30 PM
08-09-2026

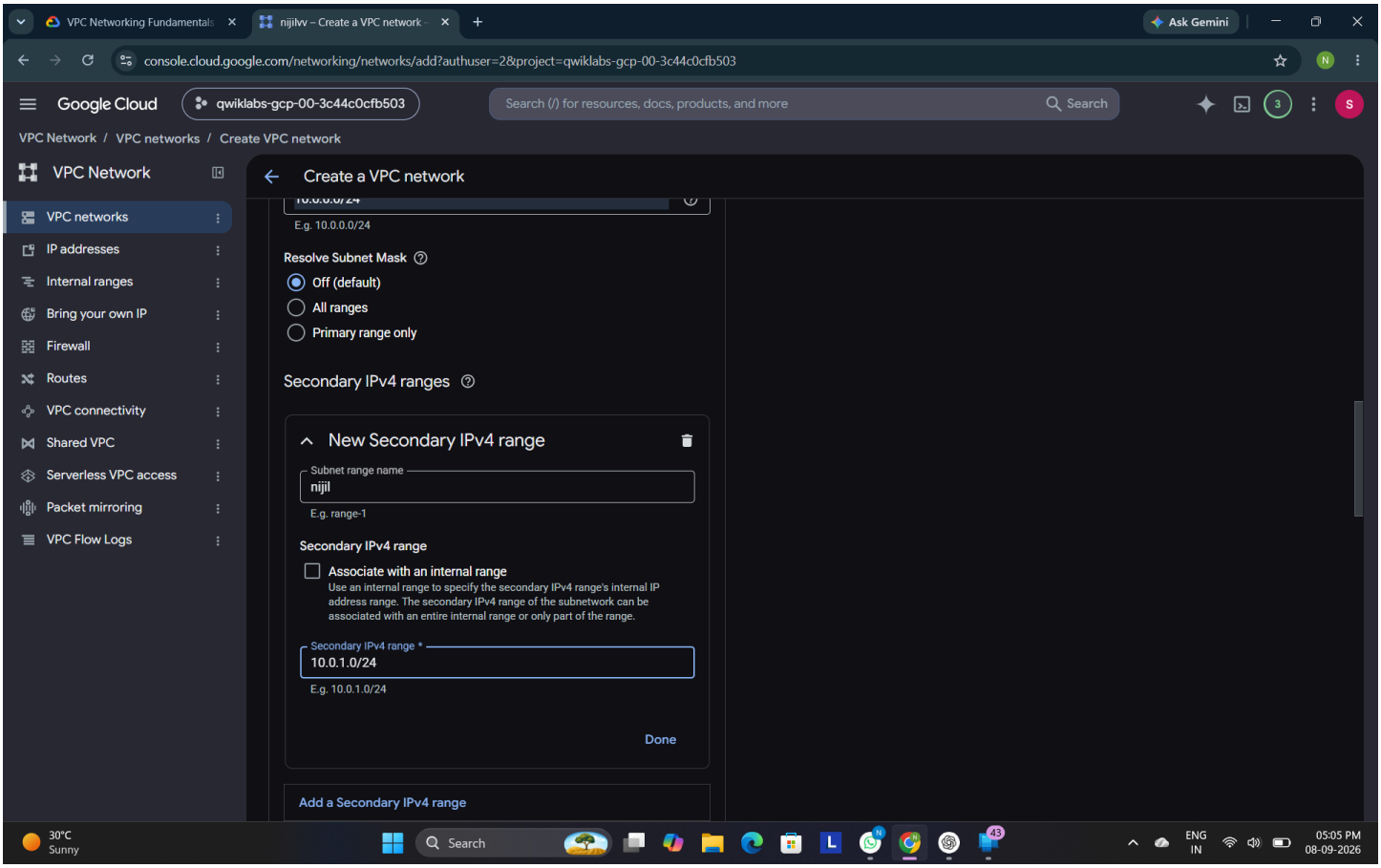

VPC Networking Fundamentals
Ask Gemini
skills.google/focuses/1229?catalog_rank=%7B"rank"%3A2%2C"num_filters"%3A0%2C"has_search"%3Atrue%7D&parent=catalog&search_id=99231829
Google Skills
What do you want to learn today?
100
0
VPC Networking Fundamentals
Lab setup and access
Use the values below to authenticate your session and configure lab resources.
Open Google Cloud console
Username
student-01-8d227e8f0417@qwi
Password
Iu9g5a386L8F
Project ID
qwiklabs-gcp-04-fde8c66af00
Browser setup
Run the lab in private browsing
Using an Incognito or private browser window is the best way to run a lab. This prevents any conflicts or extra charges to your personal account.
Dashboard
Catalog
Paths
Collections
Credentials
Subscriptions
Organizations
Lab
VPC Networking Fundamentals
1 Credit
25 minutes
Introductory
Updated 26 days ago
ID: GSP210
End
GSP210
Score: 0 / 100
59m left
This lab may incorporate AI tools to support your learning.
GSP210
Google Cloud Self-Paced Labs
30°C
Mostly sunny
Search
ENG
IN
04:31 PM
08-09-2026

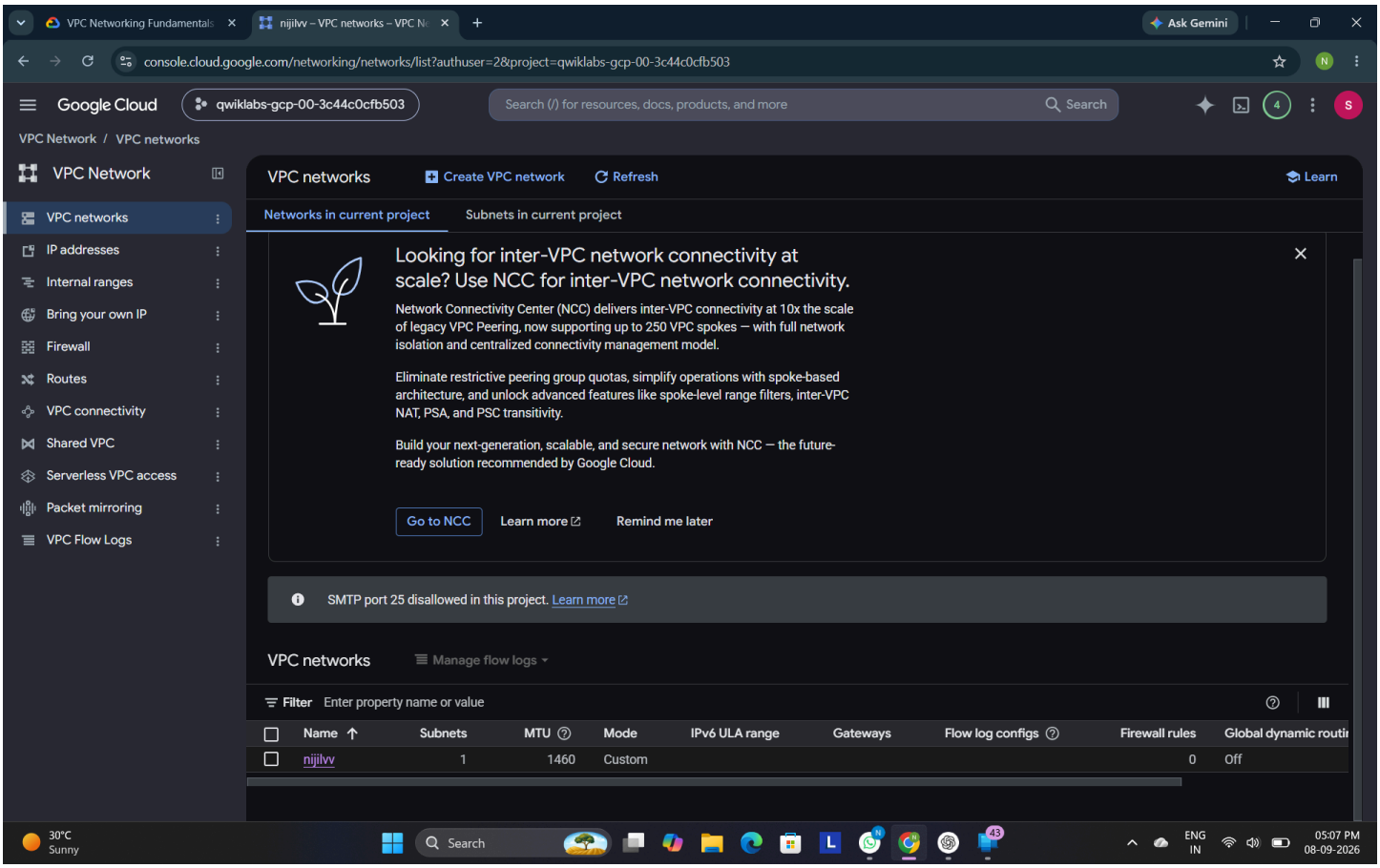

VPC Networking Fundamentals
Dashboard – qwiklabs-gcp-04-
Ask Gemini
console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-fde8c66af009&walkthrough_id=https%2F%2Fwww.skills.google%2Fdisplay_in_context%3Fdisplay_token%3DqDdSZ2pVE3zaOXSi...
Google Cloud
qwiklabs-gcp-04-fde8c66af009
Search (/) for resour...
Search
Cloud Hub
Cloud overview
Solutions
Favorite products
Favorite products appear here
Products
Billing
IAM & Admin
Marketplace
APIs & Services
Agent Platform
Compute Engine
Kubernetes Engine
Cloud Storage
Security
BigQuery
View all products
Get Agent Platform API key
Dashboard
Activity
Recommendations
Customize
Go to Cloud Hub for better visibility
This dashboard is no longer updated and will shut down in 2027. Cloud Hub offers a more comprehensive view across health, security, cost, optimization, and other data about your projects and applications.
Go to Cloud Hub
Learn more
Project info
Project name
qwiklabs-gcp-04-fde8c66af009
Project number
787059736988
Project ID
qwiklabs-gcp-04-fde8c66af009
Add people to this project
Go to project settings
APIs
Requests (requests/sec)
0.030/s
0.015/s
0
4 PM
4:30
Requests: 0.009/s
Go to APIs overview
Google Cloud Platform status
All services normal
Go to Cloud status dashboard
Billing
Estimated charges
USD $0.00
For the billing period Sep 1 – 8, 2026
Take a tour of billing
View detailed charges
Monitoring
Create my dashboard
Resources
BigQuery
Data warehouse/analytics
SQL
Managed MySQL, PostgreSQL, SQL Server
Compute Engine
LEARN
Tutorial
VPC Networking Fundamentals
GSP210
Google Cloud Self-Paced Labs
How to use this tutorial
Start
30°C
Mostly sunny
Search
ENG
IN
04:31 PM
08-09-2026

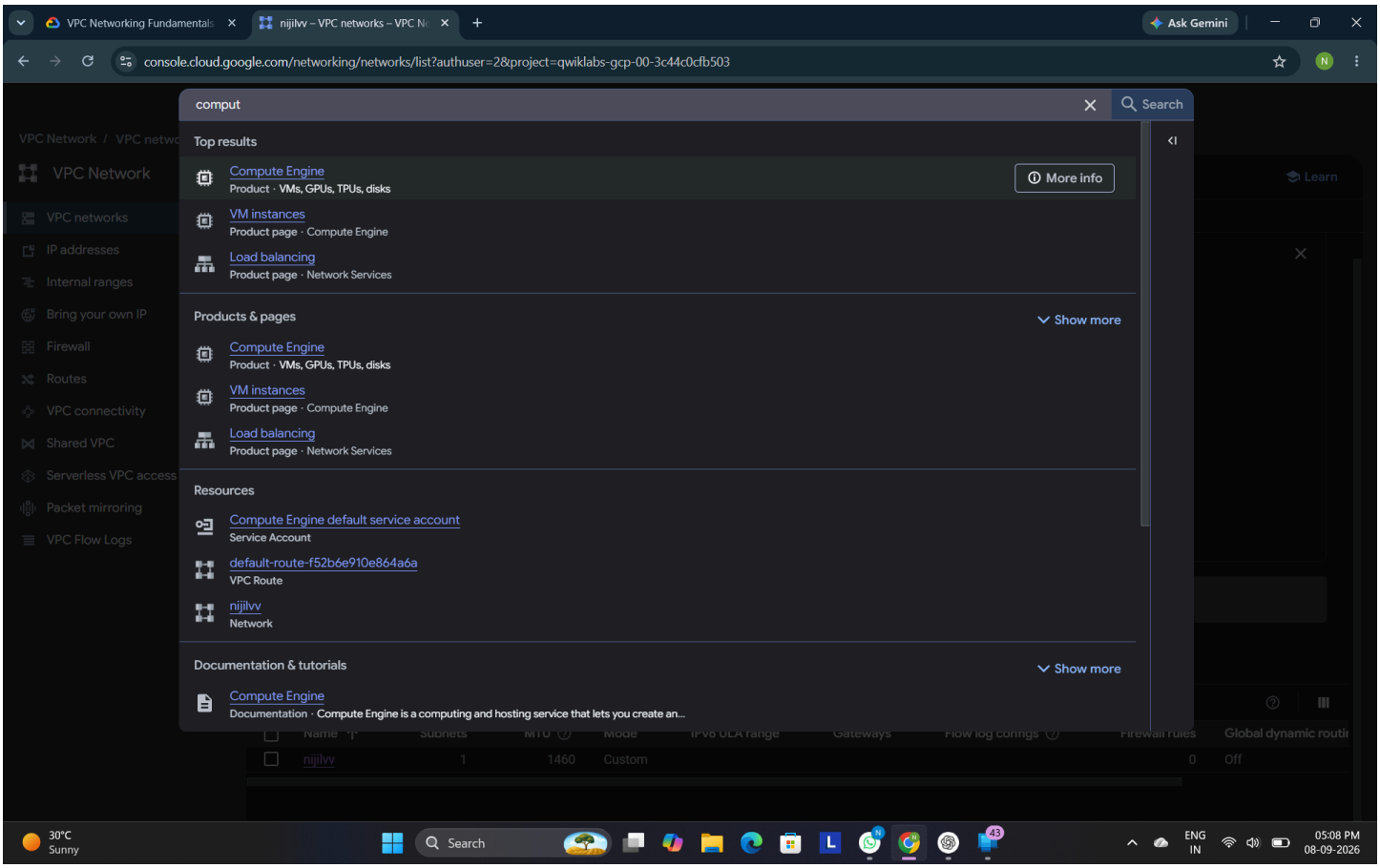

VPC Networking Fundamentals
Dashboard – qwiklabs-gcp-04-
Ask Gemini
console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-04-fde8c66af009&authuser=2&pli=1
Google Cloud
vpc
Search
Top results
VPC networks
Product page · VPC Network
More info
VPC Service Controls
Product page · Security
Firewall
Product page · VPC Network
Products & pages
Show more
VPC networks
Product page · VPC Network
VPC Service Controls
Product page · Security
Firewall
Product page · VPC Network
Documentation & tutorials
Show more
Virtual Private Cloud (VPC)
Documentation · Google Cloud VPC is global, scalable, and flexible. It provides networking fo...
VPC networks | Virtual Private Cloud
Documentation · A Virtual Private Cloud (VPC) network is a virtual version of a physical...
Test between VM instances in a VPC network
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Marketplace
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Serverless VPC Access API
Marketplace Product · Google Enterprise API · API for managing VPC access connectors.
Cloud Hub
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Marketplace
APIs & Services
Agent Platform
Compute Engine
Kubernetes Engine
Cloud Storage
Security
BigQuery
View all products
Get Agent Platform API key
Monitoring
Create my dashboard
Set up alerting policies
https://console.cloud.google.com/networking/networks/list?referrer=search&authuser=2&project=qwiklabs-gcp-04-fde8c66af009
30°C
Sunny
Search
ENG
IN
04:47 PM
08-09-2026

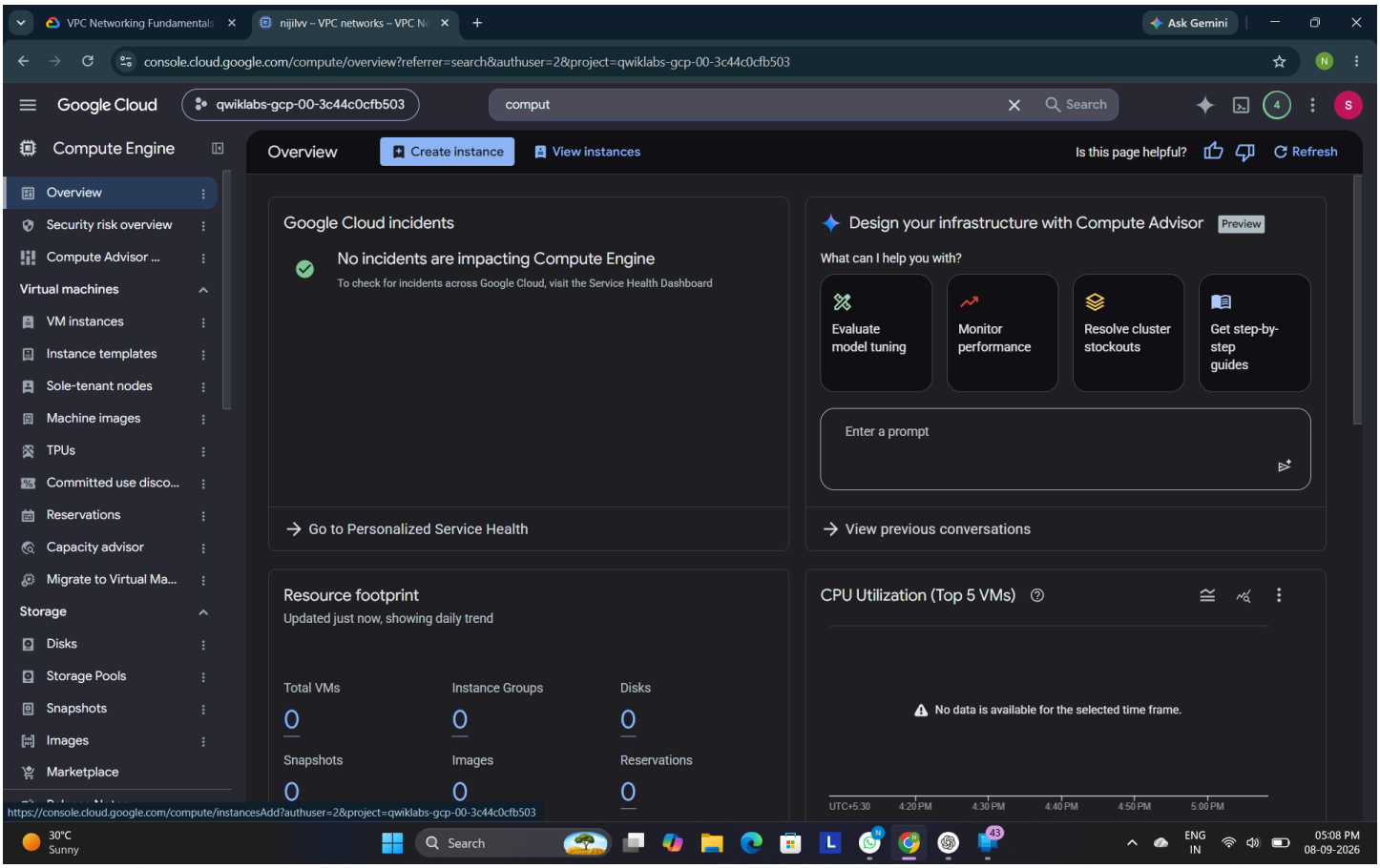

VPC Networking Fundamentals
default – VPC networks – VPC
Ask Gemini
console.cloud.google.com/networking/networks/list?authuser=2&project=qwiklabs-gcp-04-fde8c66af009
Google Cloud
qwiklabs-gcp-04-fde8c66af009
vpc
Search
VPC Network / VPC networks
VPC Network
VPC networks
Create VPC network
Refresh
Learn
Networks in current project
Subnets in current project
VPC networks
IP addresses
Internal ranges
Bring your own IP
Firewall
Routes
VPC connectivity
Shared VPC
Serverless VPC access
Packet mirroring
VPC Flow Logs
Looking for inter-VPC network connectivity at scale? Use NCC for inter-VPC network connectivity.
Network Connectivity Center (NCC) delivers inter-VPC connectivity at 10x the scale of legacy VPC Peering, now supporting up to 250 VPC spokes — with full network isolation and centralized connectivity management model.
Eliminate restrictive peering group quotas, simplify operations with spoke-based architecture, and unlock advanced features like spoke-level range filters, inter-VPC NAT, PSA, and PSC transitivity.
Build your next-generation, scalable, and secure network with NCC — the future-ready solution recommended by Google Cloud.
Go to NCC
Learn more
Remind me later
SMTP port 25 disallowed in this project. Learn more
VPC networks
Manage flow logs
Filter
Enter property name or value
Name
Subnets
MTU
Mode
IPv6 ULA range
Gateways
Flow log configs
Firewall rules
Global dynamic routi
default
30°C
Sunny
Search
ENG
IN
04:48 PM
08-09-2026

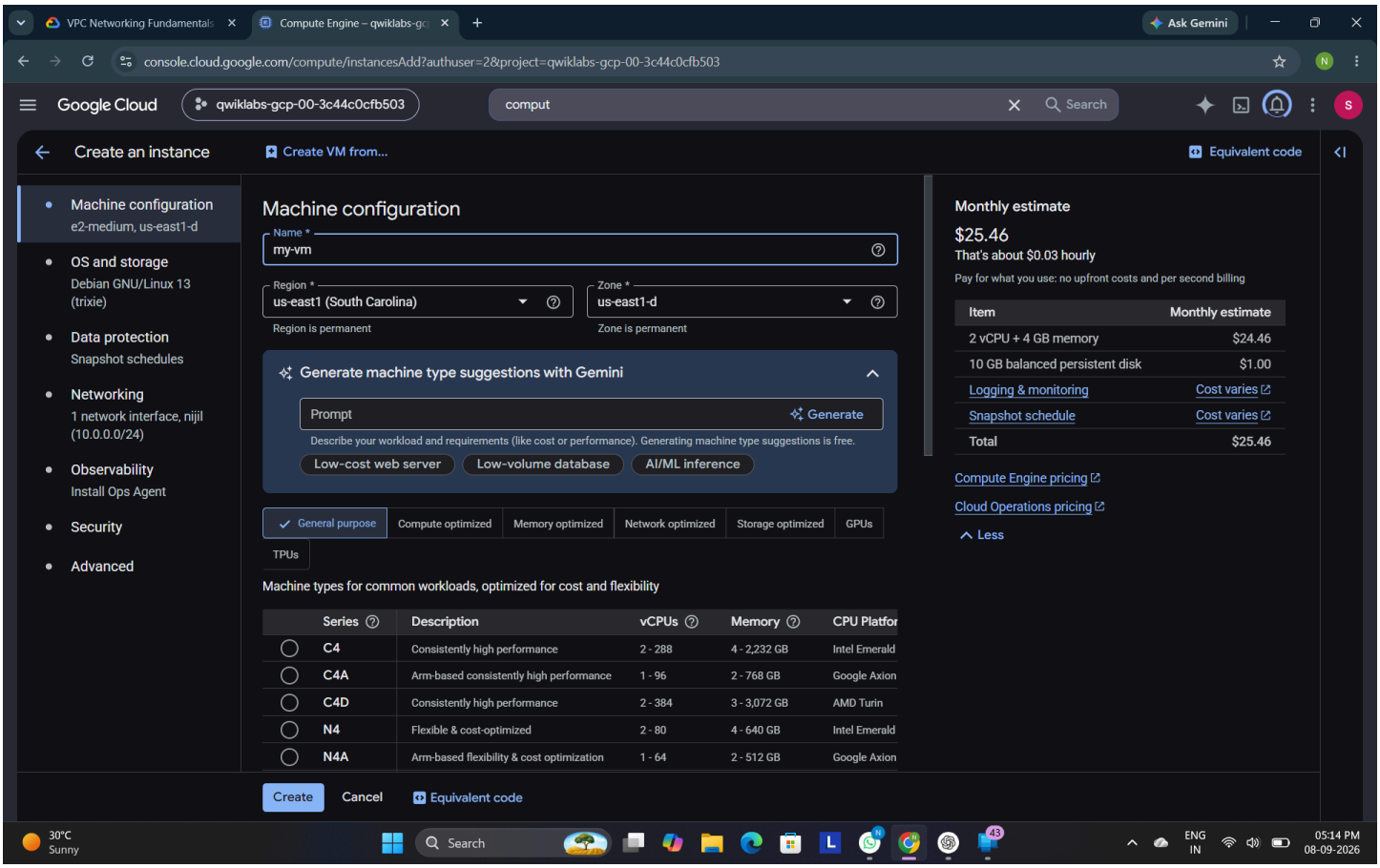

VPC Networking Fundamentals
default – Create a VPC network
Ask Gemini
console.cloud.google.com/networking/networks/add?authuser=2&project=qwiklabs-gcp-04-fde8c66af009
Google Cloud
qwiklabs-gcp-04-fde8c66af009
vpc
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VPC Network / VPC networks / Create VPC network
VPC Network
Create a VPC network
the subnets. Learn more
VPC networks
IP addresses
Internal ranges
Bring your own IP
Firewall
Routes
VPC connectivity
Shared VPC
Serverless VPC access
Packet mirroring
VPC Flow Logs
New subnet
Name *
nijil
Lowercase letters, numbers, hyphens allowed
Description
Region *
us-central1 (Iowa)
IP stack type
IPv4 (single-stack)
IPv4 and IPv6 (dual-stack)
IPv6 (single-stack)
Primary IPv4 range
Associate with an internal range
Use an internal range to specify the subnet's internal IP address range. The subnetwork can be associated with an entire internal range or only part of the range.
IPv4 range *
10.0.0.0/24
E.g. 10.0.0.0/24
Resolve Subnet Mask
Off (default)
30°C
Sunny
Search
ENG
IN
04:51 PM
08-09-2026

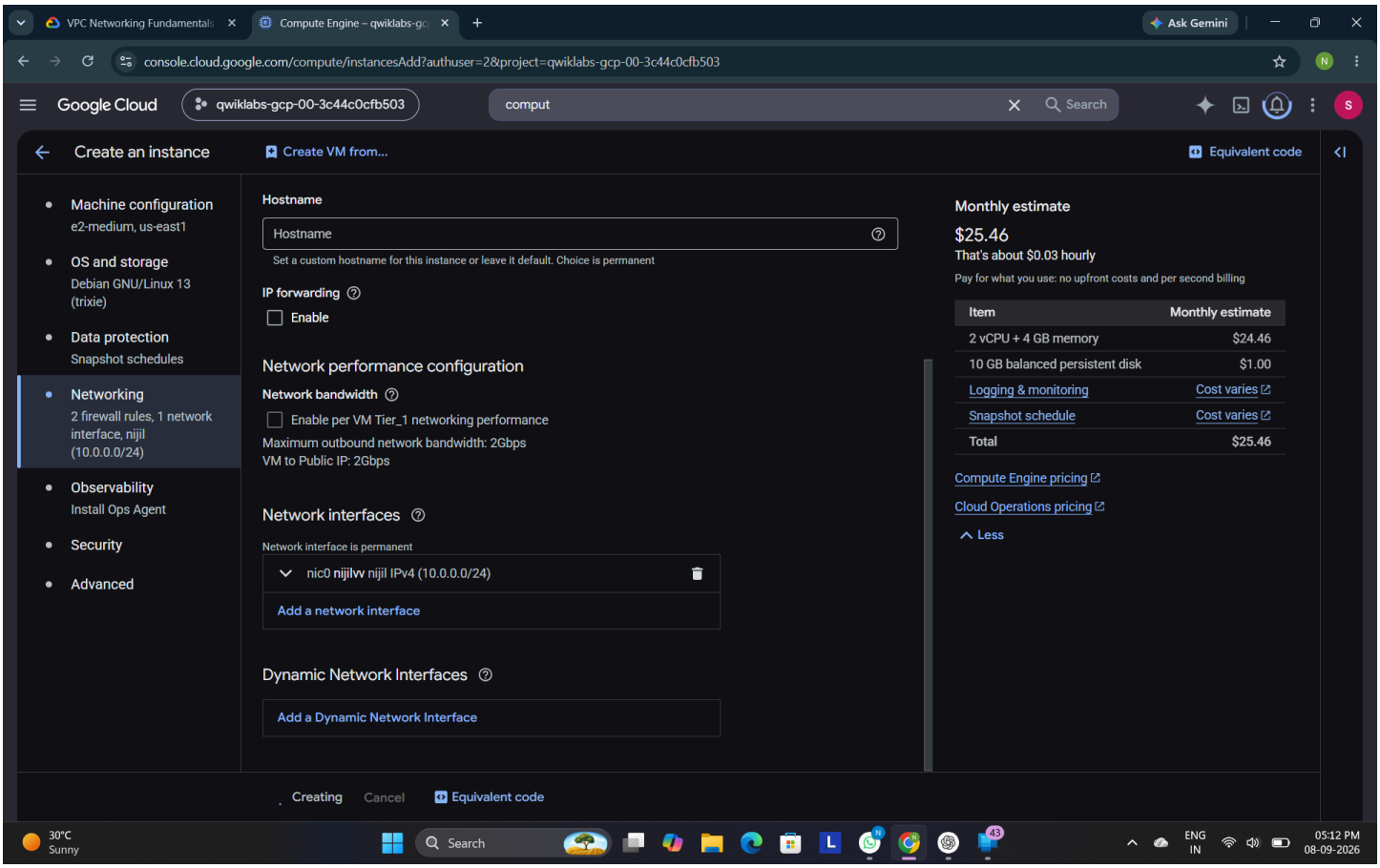

VPC Networking Fundamentals
nijilvv – Create a VPC network
Ask Gemini
console.cloud.google.com/networking/networks/add?authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
Google Cloud
qwiklabs-gcp-00-3c44c0cfb503
Search (/) for resources, docs, products, and more
Search
VPC Network / VPC networks / Create VPC network
VPC Network
VPC networks
IP addresses
Internal ranges
Bring your own IP
Firewall
Routes
VPC connectivity
Shared VPC
Serverless VPC access
Packet mirroring
VPC Flow Logs
Create a VPC network
E.g. 10.0.0.0/24
Resolve Subnet Mask
Off (default)
All ranges
Primary range only
Secondary IPv4 ranges
New Secondary IPv4 range
Subnet range name
nijil
E.g. range-1
Secondary IPv4 range
Associate with an internal range
Use an internal range to specify the secondary IPv4 range's internal IP address range. The secondary IPv4 range of the subnetwork can be associated with an entire internal range or only part of the range.
Secondary IPv4 range *
10.0.1.0/24
E.g. 10.0.1.0/24
Done
Add a Secondary IPv4 range
30°C
Sunny
Search
ENG
IN
05:05 PM
08-09-2026

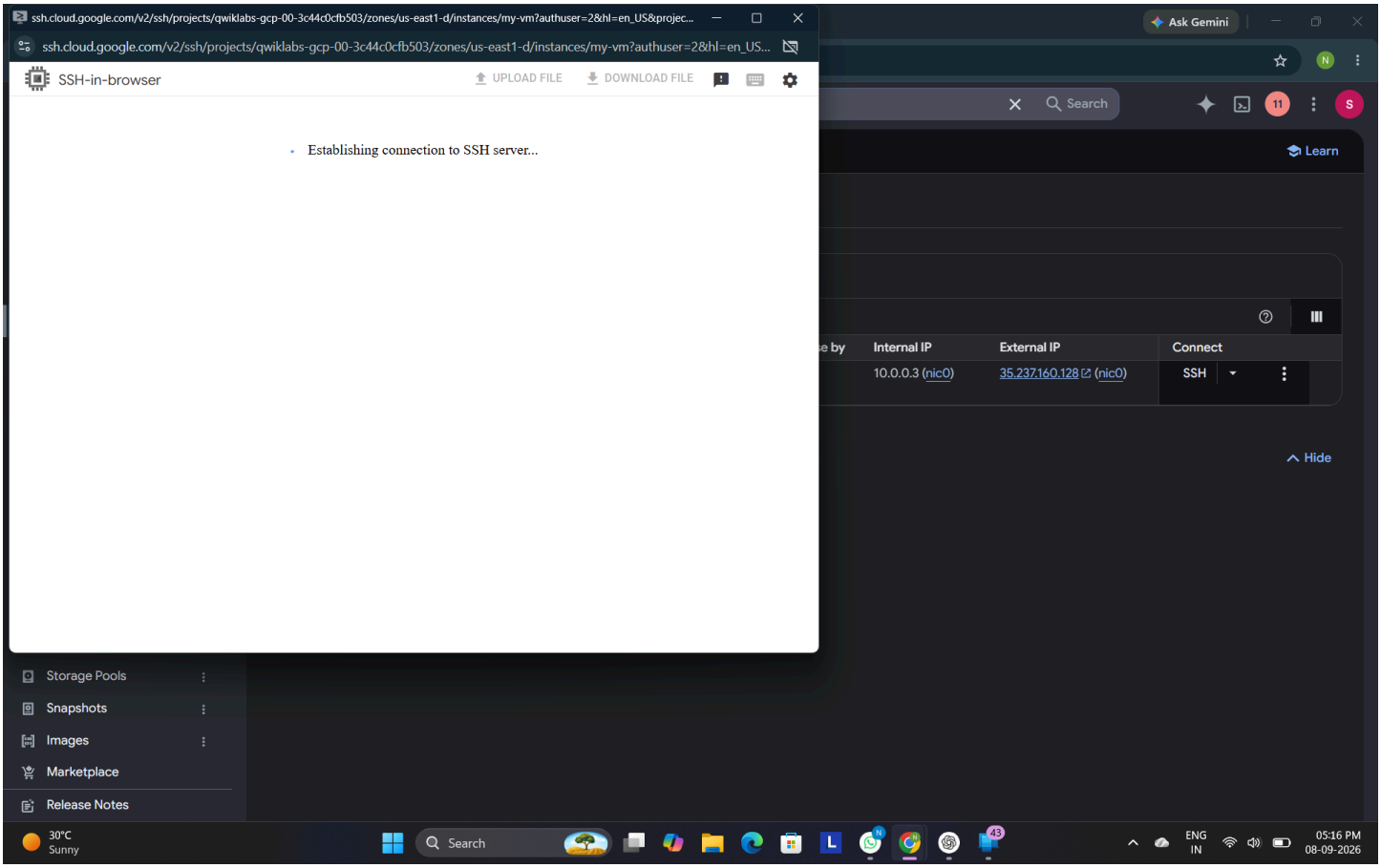

VPC Networking Fundamentals
nijilvv – VPC networks – VPC Ne
Ask Gemini
console.cloud.google.com/networking/networks/list?authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
Google Cloud
qwiklabs-gcp-00-3c44c0cfb503
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Serverless VPC access
Packet mirroring
VPC Flow Logs
VPC networks
Create VPC network
Refresh
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Subnets in current project
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Manage flow logs
Filter
Enter property name or value
Name
Subnets
MTU
Mode
IPv6 ULA range
Gateways
Flow log configs
Firewall rules
Global dynamic routi
nijilvv
1
1460
Custom
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Off
30°C
Sunny
Search
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08-09-2026


VPC Networking Fundamentals
nijilvv – VPC networks – VPC Ne
Ask Gemini
console.cloud.google.com/networking/networks/list?authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
comput
Search
Top results
Compute Engine
Product · VMs, GPUs, TPUs, disks
More info
VM instances
Product page · Compute Engine
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Product page · Network Services
Products & pages
Show more
Compute Engine
Product · VMs, GPUs, TPUs, disks
VM instances
Product page · Compute Engine
Load balancing
Product page · Network Services
Resources
Compute Engine default service account
Service Account
default-route-f52b6e910e864a6a
VPC Route
nijilvv
Network
Documentation & tutorials
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Compute Engine
Documentation · Compute Engine is a computing and hosting service that lets you create an...
VPC Network
VPC networks
IP addresses
Internal ranges
Bring your own IP
Firewall
Routes
VPC connectivity
Shared VPC
Serverless VPC access
Packet mirroring
VPC Flow Logs
Learn
nijilvv
1460
Custom
Off
30°C
Sunny
Search
ENG
IN
05:08 PM
08-09-2026


VPC Networking Fundamentals
nijilvv – VPC networks – VPC Ne
Ask Gemini
console.cloud.google.com/compute/overview?referrer=search&authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
Google Cloud
qwiklabs-gcp-00-3c44c0cfb503
comput
Search
Compute Engine
Overview
Create instance
View instances
Is this page helpful?
Refresh
Overview
Security risk overview
Compute Advisor ...
Virtual machines
VM instances
Instance templates
Sole-tenant nodes
Machine images
TPUs
Committed use disco...
Reservations
Capacity advisor
Migrate to Virtual Ma...
Storage
Disks
Storage Pools
Snapshots
Images
Marketplace
Google Cloud incidents
No incidents are impacting Compute Engine
To check for incidents across Google Cloud, visit the Service Health Dashboard
Go to Personalized Service Health
Design your infrastructure with Compute Advisor
Preview
What can I help you with?
Evaluate model tuning
Monitor performance
Resolve cluster stockouts
Get step-by-step guides
Enter a prompt
View previous conversations
Resource footprint
Updated just now, showing daily trend
Total VMs
0
Instance Groups
0
Disks
0
Snapshots
0
Images
0
Reservations
0
CPU Utilization (Top 5 VMs)
No data is available for the selected time frame.
UTC+5:30
4:20 PM
4:30 PM
4:40 PM
4:50 PM
5:00 PM
https://console.cloud.google.com/compute/instancesAdd?authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
30°C
Sunny
Search
ENG
IN
05:08 PM
08-09-2026


VPC Networking Fundamentals
Compute Engine – qwiklabs-gcp
Ask Gemini
console.cloud.google.com/compute/instancesAdd?authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
Google Cloud
qwiklabs-gcp-00-3c44c0cfb503
comput
Search
Create an instance
Create VM from...
Equivalent code
Machine configuration
e2-medium, us-east1-d
OS and storage
Debian GNU/Linux 13 (trixie)
Data protection
Snapshot schedules
Networking
1 network interface, nijil (10.0.0.0/24)
Observability
Install Ops Agent
Security
Advanced
Machine configuration
Name *
my-vm
Region *
us-east1 (South Carolina)
Region is permanent
Zone *
us-east1-d
Zone is permanent
Generate machine type suggestions with Gemini
Prompt
Generate
Describe your workload and requirements (like cost or performance). Generating machine type suggestions is free.
Low-cost web server
Low-volume database
AI/ML inference
General purpose
Compute optimized
Memory optimized
Network optimized
Storage optimized
GPUs
TPUs
Machine types for common workloads, optimized for cost and flexibility
Series
Description
vCPUs
Memory
CPU Platfor
C4
Consistently high performance
2 - 288
4 - 2,232 GB
Intel Emerald
C4A
Arm-based consistently high performance
1 - 96
2 - 768 GB
Google Axion
C4D
Consistently high performance
2 - 384
3 - 3,072 GB
AMD Turin
N4
Flexible & cost-optimized
2 - 80
4 - 640 GB
Intel Emerald
N4A
Arm-based flexibility & cost optimization
1 - 64
2 - 512 GB
Google Axion
Monthly estimate
$25.46
That's about $0.03 hourly
Pay for what you use: no upfront costs and per second billing
Item
Monthly estimate
2 vCPU + 4 GB memory
$24.46
10 GB balanced persistent disk
$1.00
Logging & monitoring
Cost varies
Snapshot schedule
Cost varies
Total
$25.46
Compute Engine pricing
Cloud Operations pricing
Less
Create
Cancel
Equivalent code
30°C
Sunny
Search
ENG
IN
05:14 PM
08-09-2026


VPC Networking Fundamentals
Compute Engine – qwiklabs-gcp
Ask Gemini
console.cloud.google.com/compute/instancesAdd?authuser=2&project=qwiklabs-gcp-00-3c44c0cfb503
Google Cloud
qwiklabs-gcp-00-3c44c0cfb503
comput
Search
Create an instance
Create VM from...
Equivalent code
Machine configuration
e2-medium, us-east1
OS and storage
Debian GNU/Linux 13 (trixie)
Data protection
Snapshot schedules
Networking
2 firewall rules, 1 network interface, nijil (10.0.0.0/24)
Observability
Install Ops Agent
Security
Advanced
Hostname
Hostname
Set a custom hostname for this instance or leave it default. Choice is permanent
IP forwarding
Enable
Network performance configuration
Network bandwidth
Enable per VM Tier_1 networking performance
Maximum outbound network bandwidth: 2Gbps
VM to Public IP: 2Gbps
Network interfaces
Network interface is permanent
nic0 nijilvv nijil IPv4 (10.0.0.0/24)
Add a network interface
Dynamic Network Interfaces
Add a Dynamic Network Interface
Monthly estimate
$25.46
That's about $0.03 hourly
Pay for what you use: no upfront costs and per second billing
Item
Monthly estimate
2 vCPU + 4 GB memory
$24.46
10 GB balanced persistent disk
$1.00
Logging & monitoring
Cost varies
Snapshot schedule
Cost varies
Total
$25.46
Compute Engine pricing
Cloud Operations pricing
Less
Creating
Cancel
Equivalent code
30°C
Sunny
Search
ENG
IN
05:12 PM
08-09-2026


ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-00-3c44c0cfb503/zones/us-east1-d/instances/my-vm?authuser=2&hl=en_US&projec...
ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-00-3c44c0cfb503/zones/us-east1-d/instances/my-vm?authuser=2&hl=en_US...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
Establishing connection to SSH server...
Ask Gemini
Search
Learn
Internal IP
External IP
Connect
10.0.0.3 (nic0)
35.237.160.128 (nic0)
SSH
Hide
Storage Pools
Snapshots
Images
Marketplace
Release Notes
30°C
Sunny
Search
ENG
IN
05:16 PM
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